

Introduction to Data Warehouse

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Topics Covered

- The Need of Data Warehousing
- Increasing Demand for Strategic Information & Inability of Past Decision Support System
- OLTP VS OLAP
- Data Warehouse Components, Benefits and Features.
- The Information flow Mechanism.
- Role of Metadata & Classification of Metadata.

The Need of Data Warehousing

Basic Understanding

Introduction

- In today's digital world, organizations such as banks, hospitals, universities, e-commerce companies, and government agencies generate enormous amounts of data every day.
- This data is stored in different operational databases (OLTP systems) that are designed to perform daily transactions efficiently. However, these systems are not suitable for analyzing large volumes of historical data or supporting strategic decision-making.
- A Data Warehouse (DW) addresses this problem by collecting data from multiple sources, integrating it into a single repository, and storing historical information in a format optimized for analysis and reporting

Introduction

- It enables managers and executives to make informed decisions based on accurate, consistent, and historical data rather than relying solely on current operational data.
- This motivation is emphasized in your PPT through the discussion of the Information Crisis, the inability of operational systems to provide strategic information, and the evolution of data warehouses.

Why do we need Data Warehousing ?

- Organizations need a data warehouse because traditional operational databases are designed to run the business, whereas a data warehouse is designed to analyze the business.
- For example:
 - A supermarket processes thousands of customer purchases every day.
 - The billing system records each transaction instantly.
 - However, the manager wants answers to questions like:
 - Which products sold the most during the last five years?
 - Which city generates the highest revenue?
 - What are customers likely to purchase together?
 - Which season produces maximum sales?
- An operational database cannot efficiently answer these analytical questions because it stores only current transactional data. A data warehouse stores historical, integrated, and summarized data, making such analysis possible.

Problems with OLTP

- a) Data is scattered across multiple systems
- b) Lack of Historical Data
- c) Poor Support for Decision Making
- d) Slow Complex Queries
- e) Inconsistent Data
- f) Difficulty in Strategic Decision Making

Information Crisis

- As organizations expanded, they experienced what is known as the Information Crisis.
- The main problems included:
 - Data scattered across many systems.
 - Different versions of the same data.
 - Difficulty locating required information.
 - Heavy dependence on IT departments.
 - Long waiting time for reports.
 - Inconsistent and poorly documented data.
- **These challenges made decision-making slow and inefficient.**

Need for Data Warehousing (Major Reasons)

- Better Decision Making
- Historical Data Analysis
- Integrated Data
- Faster Query Processing
- Trend Analysis
- Business Intelligence
- Improved Data Quality
- Enterprise-wide View
- Better Forecasting
- Competitive Advantage

Real-World Example : Amazon

- Amazon records millions of customer transactions every day.
- Operational Database:
 - Customer places an order.
 - Payment is processed.
 - Product is shipped.
- Data Warehouse:
 - Managers analyze:
 - Best-selling products.
 - Monthly sales trends.
 - Customer buying behavior.
 - Regional demand.
 - Product recommendations.
- Without a data warehouse, performing these analyses would be extremely difficult and could affect the performance of the live transaction system.

Applications

- Data warehouses are widely used in:
- Banking
- Healthcare
- Retail
- E-commerce
- Insurance
- Telecommunications
- Manufacturing
- Government
- Education
- Airlines

Increasing Demand for Strategic Information & Inability of Past Decision Support System

Base of Data Warehousing

Introduction

- As businesses expanded globally and competition increased, organizations started generating huge amounts of data from various departments such as sales, finance, marketing, human resources, production, and customer service.
- Managers were no longer interested only in daily operational reports; they required strategic information that could help them analyze business performance, identify trends, forecast future outcomes, and make long-term decisions.
- Earlier Decision Support Systems (DSS) were developed to assist managers in decision-making. However, these systems had several limitations because they relied on operational databases, isolated applications, and manually generated reports.

Introduction

- As business needs became more complex, these traditional DSSs could no longer satisfy the growing demand for strategic information, which ultimately led to the development of Data Warehousing.
- This transition is reflected in your PPT through the evolution from ad hoc reports to DSS and executive information systems, along with the increasing inability of operational systems to provide strategic information.

What is Strategic Information?

- Strategic information is information used by top-level management to make long-term business decisions that affect the future direction and success of an organization.
- Unlike operational information, strategic information focuses on analyzing historical data, identifying trends, evaluating business performance, and planning future strategies.

Characteristics of Strategic Information

- Supports long-term planning.
- Based on historical and summarized data.
- Used by senior executives and managers.
- Helps in forecasting future trends.
- Used for business growth and competitive advantage.
- Requires data from multiple departments.
- Examples of Strategic Information
 - Which product has generated the highest profit over the last five years?
 - Which region contributes the maximum revenue?
 - What is the customer retention rate?

Increasing Demand for Strategic Information

- Earlier, organizations mainly focused on processing daily business transactions. However, as businesses grew, managers began demanding more analytical information.
- Reason :
 - 1. Growing Business Competition
 - 2. Huge Growth of Data
 - 3. Need for Better Decision Making
 - 4. Long-Term Business Planning
 - 5. Customer-Centric Business
 - 6. Government and Regulatory Requirements

Evolution of Decision Support Systems

Explanation of Each Stage

1. Ad Hoc Reports

- Reports prepared whenever needed.
- Mostly manual.
- Time-consuming.

2. Special Extract Programs

- Developers wrote separate programs for every report.
- Required programming effort.

3. Small Applications

- Department-specific reporting applications.
- No integration among departments.

4. Information Centers

- Teams created reports for business users.
- Users depended heavily on IT staff.

5. Decision Support Systems (DSS)

- Provided analytical reports.
- Limited by operational databases.

6. Executive Information Systems (EIS)

- Designed specifically for senior management.
- Displayed summarized business information.

7. Data Warehouse

- Solved many DSS limitations.
- Integrated enterprise-wide historical data.
- Supported Business Intelligence, OLAP, and Data Mining.

Ad Hoc Reports



Special Extract Programs



Small Applications



Information Centers



Decision Support Systems (DSS)



Executive Information Systems (EIS)



Data Warehousing

Inability of Past Decision Support Systems

- 1. Too Many User Requests
- 2. Constantly Changing Requirements
- 3. Dependence on IT Department
- 4. Data Was Scattered
- 5. Lack of Historical Data
- 6. Slow Response Time
- 7. Inconsistent Data
- 8. Limited Analytical Capability
- 9. High Maintenance Cost
- 10. Inability to Support Strategic Decisions

Cont....

- Huge number of information requests.
- IT departments unable to respond on time.
- User requirements constantly changing.
- Users asking for increasingly detailed (spiral) information.
- Heavy dependence on IT staff.
- Company-wide reporting not supported from a single system.
- Reports required custom programs, making reporting slow and expensive

OLTP VS OLAP

Comparative Study

<u>Basis of Comparison</u>	<u>OLTP (Online Transaction Processing)</u>	<u>OLAP (Online Analytical Processing)</u>
Purpose	Supports day-to-day business transactions.	Supports analysis, reporting, and decision making.
Primary Users	Clerks, cashiers, customers, operational staff.	Managers, executives, business analysts.
Type of Data	Current and operational data.	Historical, integrated, and summarized data.
Database Design	Highly normalized (3NF) to reduce redundancy.	Denormalized (Star/Snowflake Schema) for fast analysis.
Operations Performed	Insert, Update, Delete, and simple Select operations.	Complex queries, aggregation, drill-down, roll-up, slice, and dice.
Query Type	Simple and predefined queries.	Complex and ad hoc analytical queries.
Response Time	Milliseconds to a few seconds.	Few seconds to minutes depending on data size.
Data Updates	Continuous and frequent.	Periodic (daily, weekly, monthly) through ETL process.
Data Volume	Current transactional records.	Large volume of historical data accumulated over years.
Data Integrity	Very high; maintains ACID properties.	High consistency after ETL but focuses on analysis rather than transactions.
Redundancy	Very low due to normalization.	Higher due to denormalization for better query performance.
Performance Optimization	Optimized for fast transaction processing.	Optimized for fast analytical processing.
Backup Requirement	Frequent backup to prevent transaction loss.	Scheduled backup since updates are infrequent.
Examples of Queries	"What is the balance of Account No. 12345?"	"What is the yearly profit trend of all branches over the last 10 years?"
Examples of Applications	Banking systems, ATM, Railway reservation, E-commerce checkout, Payroll systems.	Business Intelligence, Sales analysis, Data Mining, Executive Dashboards, Forecasting.

KEY DIFFERENCES

<u>OLTP</u>	<u>OLAP</u>
Transaction-oriented	Analysis-oriented
Current data	Historical data
Frequent updates	Rare updates
Normalized database	Denormalized database
Fast transaction processing	Fast analytical processing
Supports daily operations	Supports strategic decisions
Many concurrent users	Comparatively fewer users
Short, simple queries	Long, complex queries

Example

OLTP Example

A customer purchases a mobile phone from an online shopping website.

- The order is placed.
- Payment is processed.
- Inventory is updated.
- Invoice is generated.

These are **transactional operations**, so they are handled by **OLTP**.

OLAP Example

The company's CEO wants to analyze:

- Which mobile brand sold the most in the last 5 years?
- Which state generated the highest revenue?
- What is the seasonal sales trend?

These are **analytical operations**, so they are handled by **OLAP**.

Exam Tip (Very Important)

Remember this simple trick:

OLTP

OLAP

Run the Business

Analyze the Business

Data Warehouse Components, Benefits and Features

How DW Separates ?

Data Warehouse Components

- A Data Warehouse is not just a database; it is a complete environment consisting of several interconnected components that collect, clean, store, manage, and analyze data from multiple sources. Each component plays an important role in transforming raw operational data into meaningful information for business decision-making.
- Data Warehousing Environment consists of four major components:
 - Data Extraction
 - Data Management
 - Data Access
 - Administration & Management

Main Components of a Data Warehouse

- 1. Data Sources
- Definition: Data sources are the origin of data that is loaded into the data warehouse.
- Sources may include:
 - Operational Databases (OLTP)
 - ERP Systems
 - CRM Systems
 - Legacy Systems
 - Excel Files
 - Web Applications
 - External Data Sources
- Example:
 - A bank collects data from:
 - Savings Account System
 - Loan Management System
 - ATM Transactions
 - Credit Card System

- 2. Data Extraction (ETL)
- ETL stands for:
- E – Extract
- T – Transform
- L – Load
- It is responsible for moving data from operational systems into the warehouse.
- a) Extract : Collects data from different sources.
- b) Transform : Removes duplicate data
 - Corrects errors
 - Converts formats
 - Integrates data
 - Standardizes values
- c) Load : Stores the cleaned data into the Data Warehouse.

- 3. Data Cleaning and Integration
 - Before loading, data must be cleaned.
 - It includes:
 - Removing duplicate records
 - Filling missing values
 - Correcting inconsistent data
 - Standardizing formats
 - Resolving naming conflicts
- 4. Data Warehouse Repository
 - This is the central storage area.
 - It stores:
 - Historical Data
 - Integrated Data
 - Subject-Oriented Data
 - Non-Volatile Data
 - This repository is optimized for analytical queries rather than transaction processing.

- 5. Metadata Repository
- Metadata means: "Data about Data."
- It contains information like:
 - Source of data
 - Data definitions
 - Data types
 - ETL rules
 - Table structures
 - Relationships
 - Loading schedules
- 6. Data Marts
 - A Data Mart is a subset of the Data Warehouse created for a specific department.
 - Examples:
 - Sales Data Mart
 - Finance Data Mart
 - HR Data Mart
 - Marketing Data Mart
 - NOTE : data marts are smaller, single-subject repositories and are often considered building blocks of a data warehouse

- 7. OLAP Server
- OLAP provides multidimensional analysis.
- It allows users to:
 - Drill Down
 - Roll Up
 - Slice
 - Dice
 - Pivot
- Managers use OLAP for business analysis.
- 8. Front-End Tools
- These are the tools used by end users.
- Examples:
 - Dashboards
 - Reports
 - Charts
 - Graphs
 - Data Mining Tools
 - Business Intelligence Tools

Data Warehouse Components Diagram



Features (Characteristics) of Data Warehouse

- According to Bill Inmon, a Data Warehouse has four fundamental characteristics: Subject-Oriented, Integrated, Time-Variant, and Non-Volatile. Your PPT also presents these as the defining features of a data warehouse
- 1. Subject-Oriented
- 2. Integrated
- 3. Time-Variant
- 4. Non-Volatile

Additional Features

- Multiple Source Integration
- Historical Data Storage
- Fast Query Processing
- Read-Oriented
- High Data Quality
- Supports Business Intelligence
- Scalability

Benefits of Data Warehouse

- Better Decision Making
- Integrated Data
- Historical Analysis
- Faster Report Generation
- Improved Data Quality
- Business Intelligence Support
- Trend Analysis
- Better Customer Understanding
- Enterprise-wide View
- Competitive Advantage

The Information flow Mechanism

Working

Introduction

- The Information Flow Mechanism in a Data Warehouse describes the complete process through which raw data is collected from various operational systems, transformed into meaningful information, stored in a centralized repository, and finally delivered to decision-makers.
- It represents the journey of data from its original sources to the end users. This mechanism ensures that data is accurate, consistent, integrated, and available for analysis and business intelligence.
- **the Data Warehousing Environment consists of Data Extraction, Data Management, Data Access, and Administration & Management, which together form the information flow mechanism.**

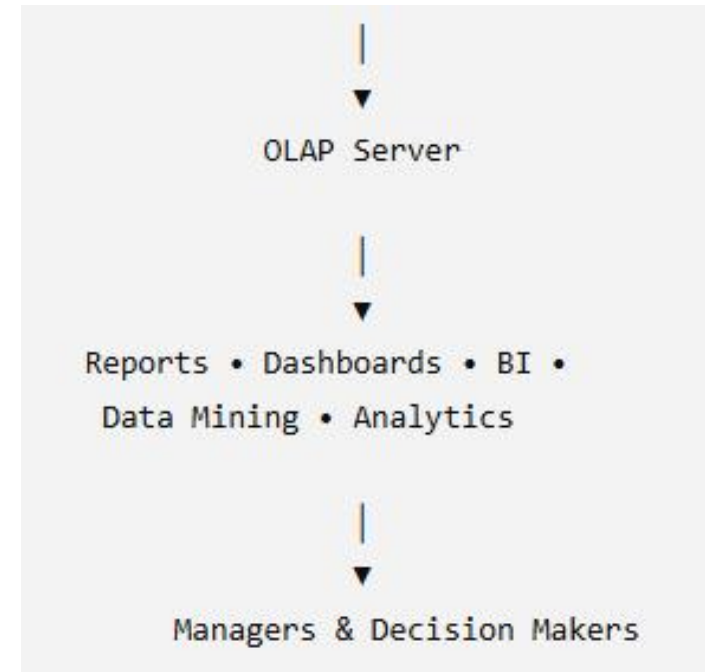
Definition

- Information Flow Mechanism is the systematic movement of data from various operational and external sources through extraction, transformation, loading, storage, and analysis until it becomes useful information for business decision-making.
- In simple words,
- Raw Data → Processed Data → Information → Knowledge → Decision

Need for Information Flow Mechanism

- Organizations generate data from many different systems.
- Examples include:
 - Sales System
 - HR System
 - Finance System
 - Inventory System
 - CRM
 - ERP
- External Data Sources
- If this data is analyzed directly,
 - Duplicate records may exist.
 - Different formats may be used.
 - Data may be incomplete.
 - Results may be inconsistent.

FLOW



Steps in the Information Flow Mechanism

- Step 1: Data Sources

- This is the starting point.
- Data is collected from different operational systems.
- Examples:
 - Banking Software
 - Hospital Management System
 - University ERP
 - E-commerce Website
 - Payroll System
- Characteristics:
 - Current data
 - Transaction-oriented
 - Distributed across departments

Step 2: Data Extraction

Data is extracted from all source systems.

Purpose:

- Collect data from multiple databases.
- Gather structured and unstructured data.
- Capture only required information.

Example:

A bank extracts

- Customer records
- Loan information
- ATM transactions
- Credit card transactions

Steps in the Information Flow Mechanism

- Step 3: Data Cleaning

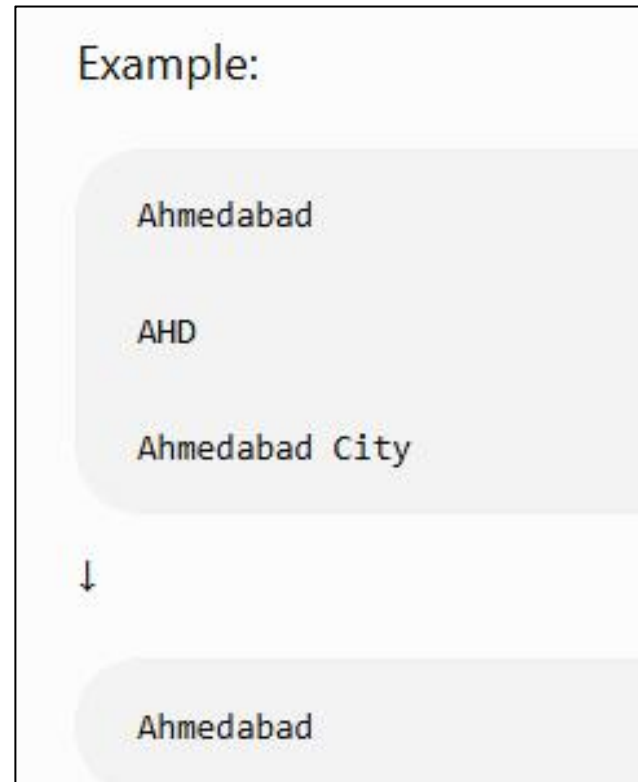
- Extracted data is rarely perfect.

- It may contain:

- Missing values
- Duplicate records
- Incorrect values
- Inconsistent formats

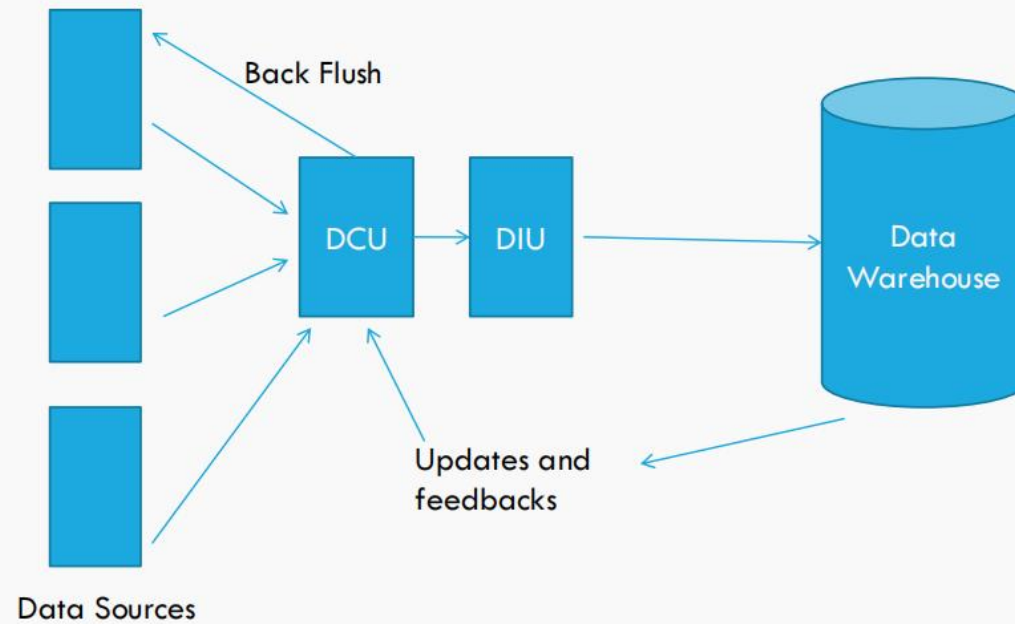
- Cleaning activities include:

- Removing duplicates
- Filling missing values
- Correcting spelling mistakes
- Removing invalid records



Daigram

DATA CLEANING AND INTEGRATION



Steps in the Information Flow Mechanism

- **Step 4: Data Transformation**

- Data from different sources must be converted into a common format.
- Transformation includes:
 - Standardization
 - Data validation
 - Data integration
 - Summarization
 - Time handling
 - Denormalization
 - Creating Fact Tables
 - Creating Dimension Tables

Steps in the Information Flow Mechanism

- Data from different departments is combined into one unified repository.

Example

Sales Database

Finance Database

Inventory Database

HR Database



Integrated Data Warehouse

Benefits:

- Eliminates redundancy.
- Provides a single version of truth.
- Enables enterprise-wide analysis.

Steps in the Information Flow Mechanism

- Step 6: Data Loading
 - After cleaning and transformation,
 - the processed data is loaded into the Data Warehouse.
 - Loading can be:
 - Initial Load
 - Entire database is loaded for the first time.
 - Incremental Load
 - Only newly added or modified records are loaded.
 - Scheduled Refresh
 - Data may be refreshed:
 - Daily
 - Weekly
 - Monthly
 - Quarterly
 - Yearly

Steps in the Information Flow Mechanism

- Step 7: Data Storage
- The Data Warehouse stores
 - Historical data
 - Subject-oriented data
 - Integrated data
 - Time-variant data
 - Non-volatile data

Step 8: Metadata Management

Metadata stores information about the stored data.

It contains :

- Table names
- Column names
- Data definitions
- Source systems
- ETL rules
- Relationships

Metadata helps users understand and access warehouse data efficiently.

Steps in the Information Flow Mechanism

- Step 9: Data Access
- End users access the warehouse through:
 - Reports
 - Dashboards
 - OLAP
 - Data Mining Tools
 - Business Intelligence Software
- Users can perform
 - Trend analysis
 - Forecasting
 - Performance analysis
 - Comparative analysis

Step 10: Decision Making

The final stage is decision making.

Managers use analytical information to answer questions such as:

- Which product generates maximum profit?
- Which branch performs best?
- Which customer segment is most profitable?
- What will next year's sales be?

Summary Diagram

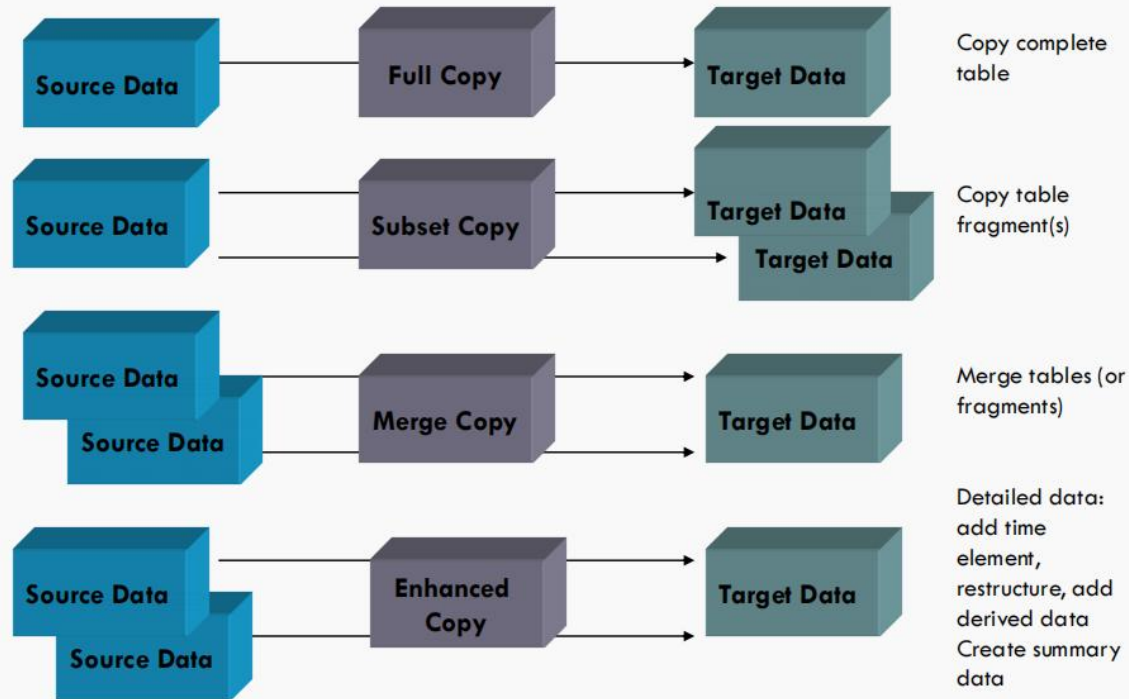


Advantages of Information Flow Mechanism

- Integrates data from multiple sources.
- Improves data quality.
- Eliminates duplicate records.
- Stores historical information.
- Supports Business Intelligence.
- Enables fast analytical queries.
- Provides consistent information.
- Improves decision making.
- Supports forecasting.
- Increases organizational efficiency.

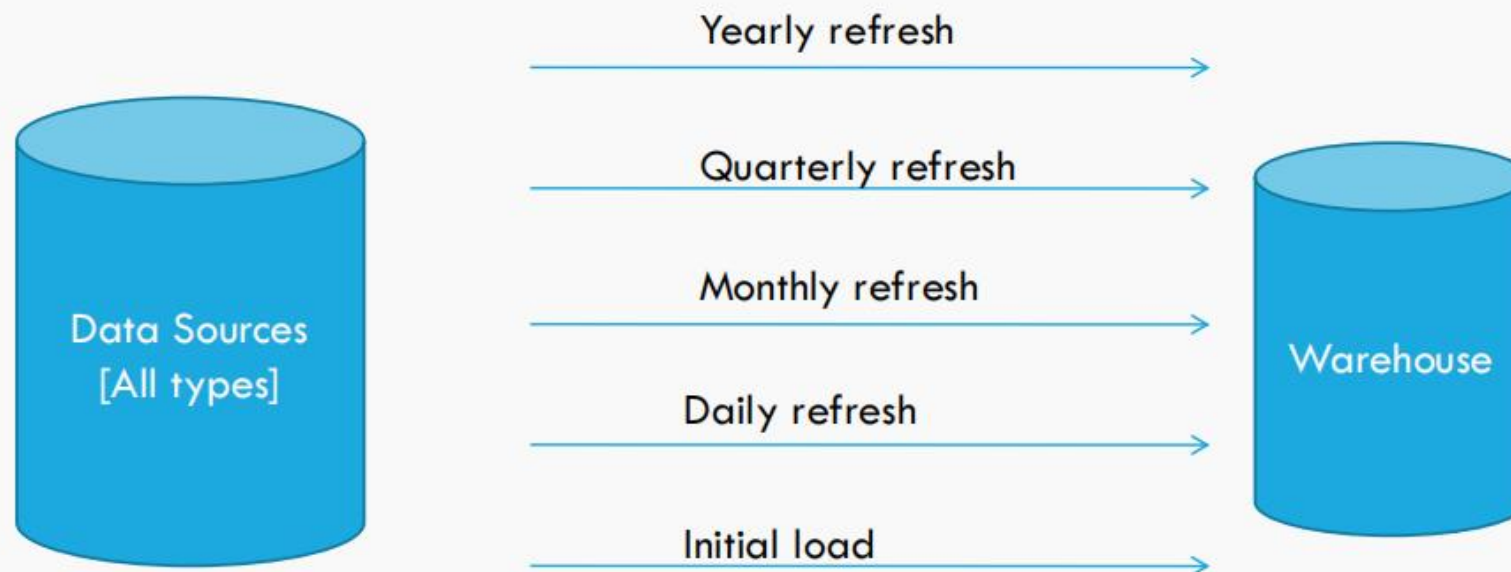
Additional Diagram

LOADING AND SCHEDULING



Data Movements to Warehouse

DATA MOVEMENTS TO WAREHOUSE



This leads to the time variant characteristic of data warehouse

Role of Metadata & Classification of Metadata

Data About Data!

Introduction

- A Data Warehouse stores a huge amount of integrated and historical data collected from multiple sources. However, simply storing data is not enough. Users also need to know what the data means, where it came from, how it was transformed, when it was loaded, and how it should be used.
- This information is called Metadata.
- Metadata acts as a guide or directory for the Data Warehouse. It helps developers, administrators, and business users understand the structure, meaning, and origin of the data, making the warehouse easier to manage and use.
- The importance of metadata is also emphasized in your PPT, where Metadata for Sources is listed during data extraction and Metadata Creation is included as part of the transformation process before loading the warehouse.

Definition of Metadata

- Metadata is "Data about Data."
- It provides descriptive information about the data stored in the Data Warehouse.
- Simple Definition :
- Metadata is information that describes the structure, meaning, source, format, transformation, and usage of the data stored in a Data Warehouse.

Need of MetaData

- Imagine a Data Warehouse containing thousands of tables and millions of records.
- Without metadata,
 - Users would not know what each table represents.
 - Developers would not know where the data originated.
 - ETL developers would not know how the data was transformed.
 - Reports could become inconsistent.
- Metadata provides all this information in one place.

Role of MetaData in DW!

- Metadata plays a vital role throughout the Data Warehouse lifecycle.
- 1. Describes the Data
 - Metadata explains:
 - Table names
 - Column names
 - Data types
 - Field descriptions
 - Constraints

Example:

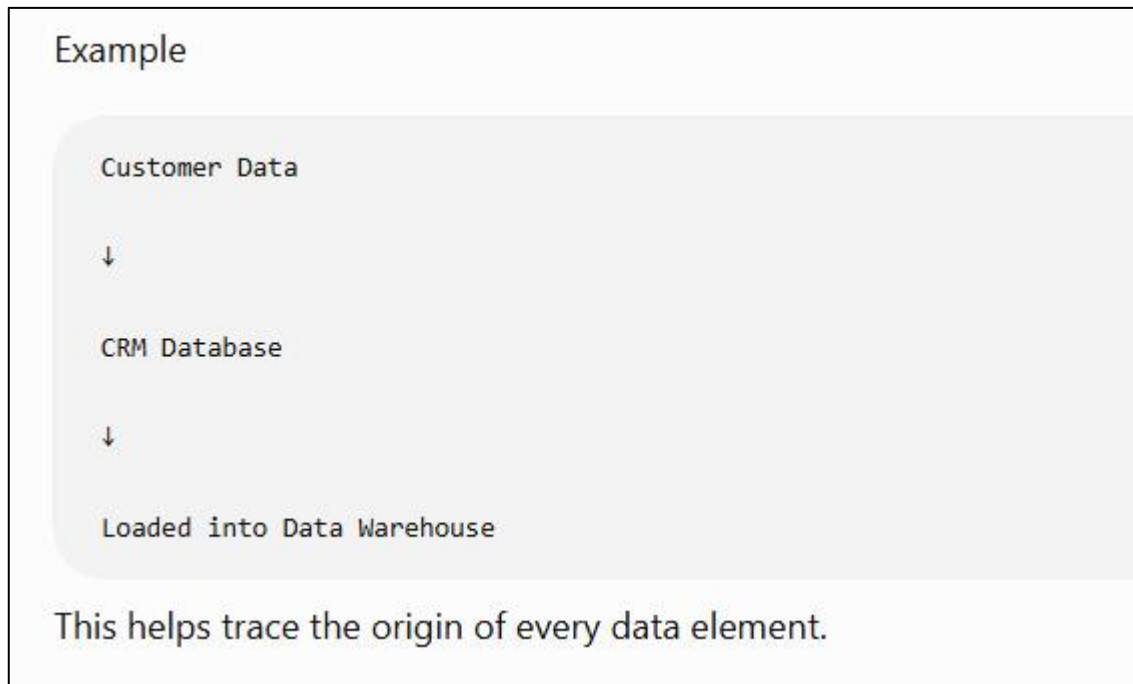
Column : Customer_Name

Data Type : VARCHAR(50)

Description : Name of Customer

2. Identifies Data Sources

- Metadata records where the data originated.



3. Supports ETL Process

- Metadata stores:
 - Extraction rules
 - Transformation rules
 - Loading schedules
- Example:
 - Remove duplicate customers.
 - Convert Date format to YYYY-MM-DD.
 - Replace NULL values with "Unknown".

4. Improves Data Integration

- Since data comes from multiple systems,
- metadata ensures:
- Common naming conventions.
- Uniform data formats.
- Consistent units.
- Standard codes.

Example:

M

Male

1

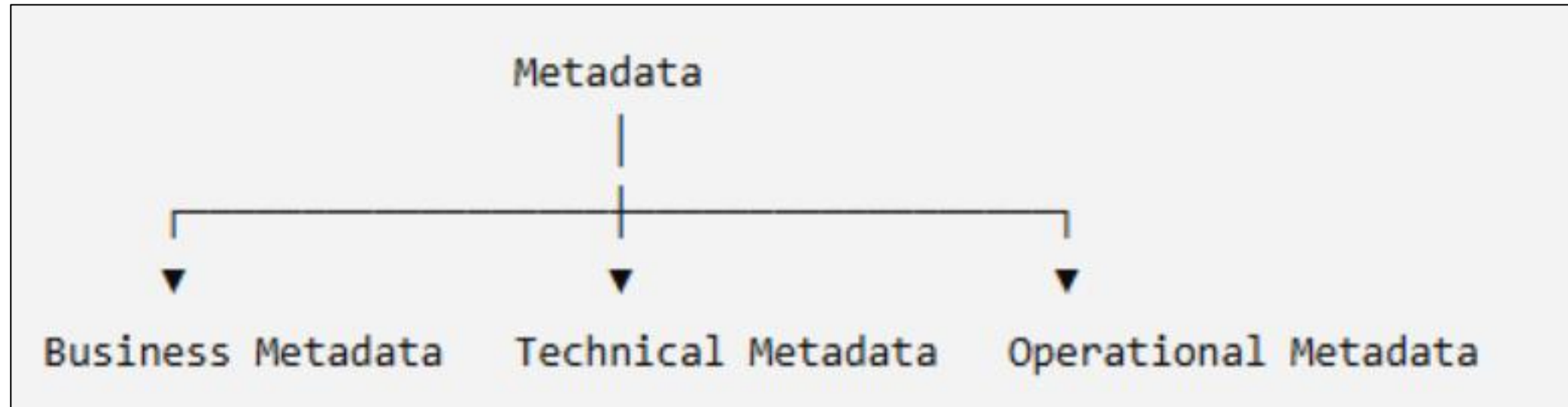
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Male

Other Roles of MetaData

- 5. Helps End Users
- 6. Improves Query Performance
- 7. Supports Data Lineage
 - Metadata records the complete journey of data.
- 8. Simplifies Maintenance
- 9. Enhances Security
- 10. Supports Business Intelligence

Classification (Types) of Metadata



1. Business Metadata

- Business Metadata describes data in business language that is easy for managers and end users to understand.
- It answers:
 - What does this data mean?
 - Why is it stored?
 - How is it used in business?
- Contains
 - Business definitions
 - Business rules
 - Report descriptions
 - KPIs
 - Policies
- Users
 - Managers
 - Executives
 - Business Analysts

Example

Field : Annual Sales

Meaning : Total Sales of One Financial Year

2. Technical Metadata

- Technical Metadata describes the technical structure of the Data Warehouse.
- It is mainly used by developers and database administrators.
- Contains
 - Table names
 - Column names
 - Data types
 - Keys
 - Relationships
 - Schemas
 - ETL mappings
 - Indexes
- Users
 - Database Administrators
 - Developers
 - ETL Engineers

Example

Table : Customer

Primary Key : Customer_ID

Data Type : Integer

3. Operational Metadata

- Operational Metadata describes the execution and management of Data Warehouse operations.
- It tells:
 - When data was loaded.
 - Whether ETL completed successfully.
 - Number of records processed.
 - Processing time.
- Contains
 - ETL execution logs
 - Loading schedules
 - Refresh frequency
 - Error logs
 - Backup history
 - Data quality reports
- Users
 - System Administrators
 - ETL Developers
 - Data Warehouse Administrators

Example

Last Refresh

01 July 2026

Time

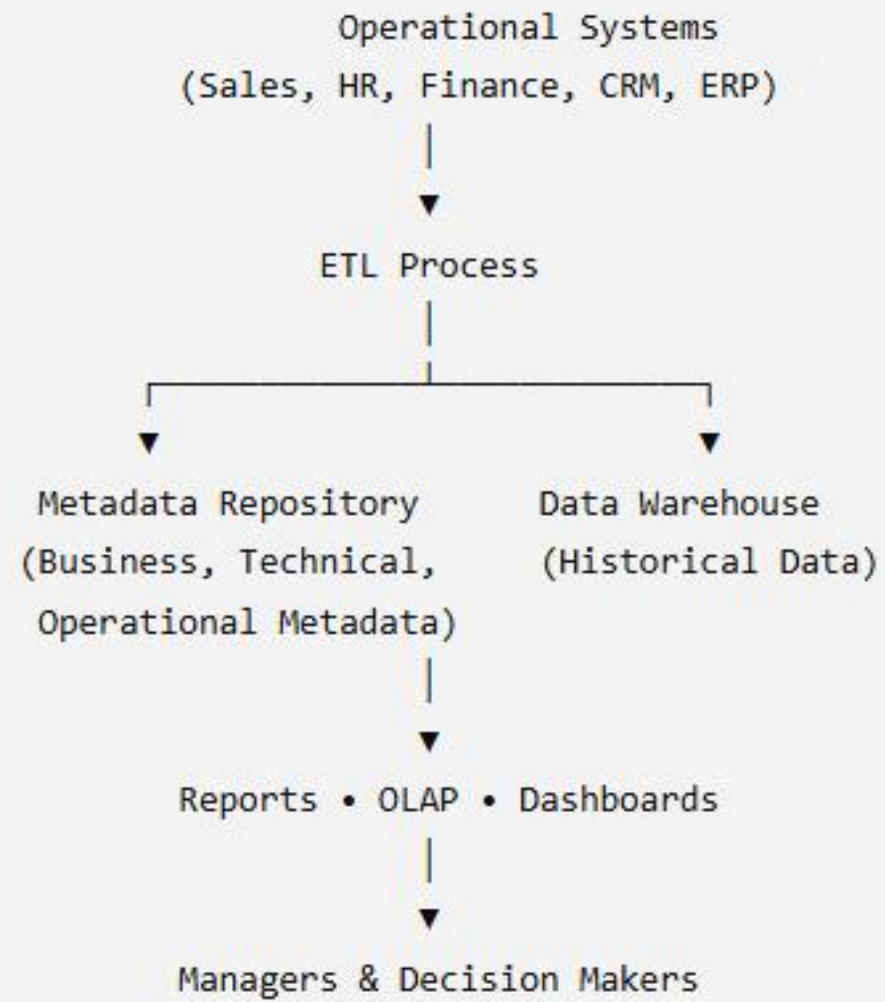
2:00 AM

Records Loaded

5,40,000

Advantages

- Makes data easy to understand.
- Improves data consistency.
- Supports ETL processes.
- Helps users locate required data.
- Improves data quality.
- Simplifies warehouse maintenance.
- Enhances security and governance.
- Supports Business Intelligence tools.
- Tracks the origin and movement of data (data lineage).
- Improves query performance.



DW & Data Marts

Smaller Subset of DW - Data Warehouse

Introduction

- As organizations grow, they generate massive amounts of data from different departments such as Sales, Finance, Human Resources (HR), Marketing, Inventory, Customer Relationship Management (CRM), and Production.
- Managing and analyzing all this data through a single operational database becomes difficult.
- To support strategic decision-making, organizations use Data Warehouses (DW) and Data Marts.
- A Data Warehouse is a centralized repository that stores integrated, historical, and subject-oriented data collected from the entire organization.
- In contrast, a Data Mart is a smaller subset of a Data Warehouse that is designed to serve the analytical needs of a specific department or business function.

Introduction

- The concept of Data Marts was introduced to simplify implementation, reduce development time, and provide faster access to departmental information while still supporting enterprise-wide decision making.
- Data Marts are smaller, subject-specific repositories, advises organizations to avoid "boiling the ocean" (building everything at once), and compares enterprise Data Warehouses with departmental Data Marts

What is Data Warehouse ?

- According to Bill Inmon,
- A Data Warehouse is a subject-oriented, integrated, time-variant, and non-volatile collection of data that supports management's decision-making process.
- A Data Warehouse integrates data from multiple operational systems into a single repository for business analysis.

Characteristics of Data Warehouse

- Subject-Oriented
- Integrated
- Time-Variant
- Non-Volatile
- Historical Data
- Enterprise-wide Repository
- Read-Oriented
- Supports OLAP and Business Intelligence

Example

A multinational bank collects data from:

- Savings Accounts
- Loan Systems
- Credit Cards
- Internet Banking
- ATM Transactions

Instead of analyzing these systems separately, all data is stored in a single Data Warehouse.

What is Data Mart ?

- A Data Mart is a small, subject-oriented subset of a Data Warehouse that focuses on a specific department or business function.
- It contains only the data required by a particular group of users.
- Characteristics :
 - Department-oriented
 - Small in size
 - Faster implementation
 - Lower development cost
 - Limited scope
 - Supports departmental decision-making

Example : Sales Department

Needs only:

- Product Sales
- Customer Orders
- Revenue
- Discounts

A Sales Data Mart stores only this information.

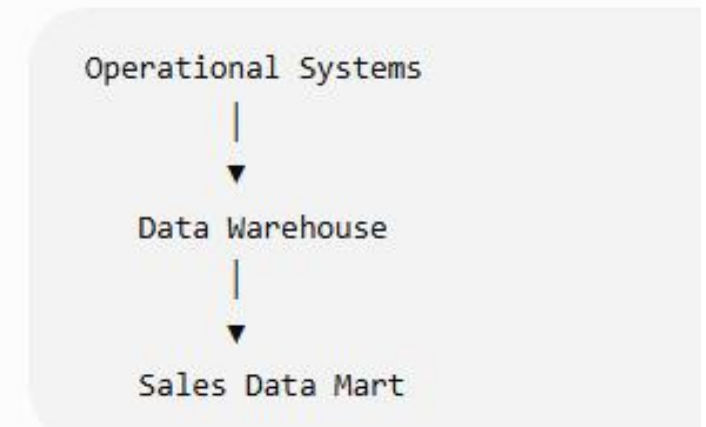
Need of Datamart

- Building a complete enterprise Data Warehouse can be:
 - Expensive
 - Time-consuming
 - Complex
- Instead, organizations often develop departmental Data Marts first.
- Advantages include:
 - Faster deployment
 - Reduced project risk
 - Lower implementation cost
 - Immediate business value

Types of DataMart : there are three major types of Data Marts!

- 1. Dependent Data Mart : A Dependent Data Mart is created from an existing Data Warehouse.
- Features
 - Receives data from the Data Warehouse.
 - Data is already cleaned and integrated.
 - Maintains consistency across the organization.
- Advantages
 - High data consistency
 - Easy maintenance
 - Enterprise-wide standardization

Architecture



2. Independent Data Mart

- An Independent Data Mart is built directly from operational databases without using a central Data Warehouse.
- Advantages
 - Faster implementation
 - Lower initial cost
- Disadvantages
 - Data duplication
 - Inconsistent information
 - Difficult integration

Architecture

Operational Database

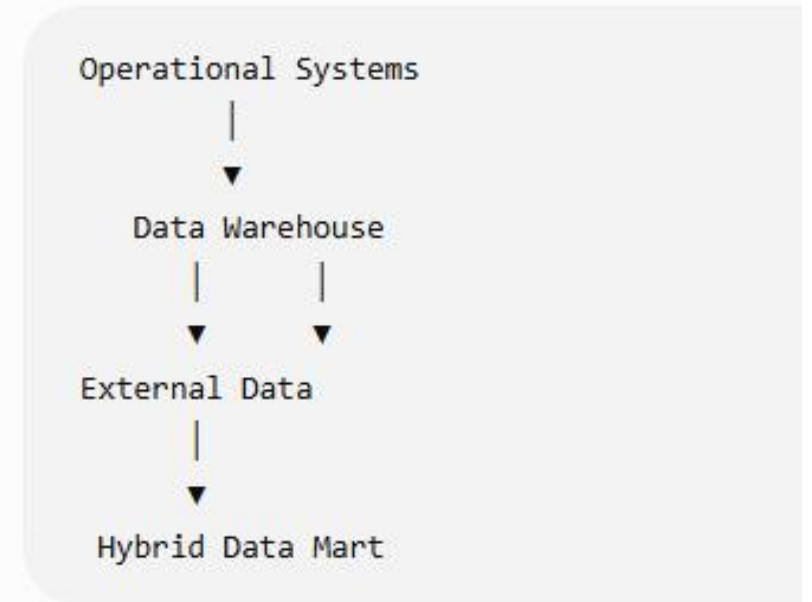


Marketing Data Mart

3. Hybrid Data Mart

- A Hybrid Data Mart receives data from both:
 - Data Warehouse
 - Operational Systems
- Advantage
 - Provides greater flexibility while maintaining consistency.

Architecture



Difference

Basis	Data Warehouse	Data Mart
Definition	Central repository for the entire organization	Subset of a Data Warehouse for a specific department
Scope	Enterprise-wide	Department-specific
Users	Entire organization	Individual department
Data Coverage	Multiple business subjects	Single business subject
Size	Very large (GB to TB or PB)	Small (GB level)
Data Source	Multiple operational systems	Data Warehouse or operational systems
Implementation Time	Long	Short
Cost	High	Low
Complexity	High	Relatively low
Maintenance	More complex	Easier
Performance	Handles enterprise-wide analysis	Faster departmental queries
Examples	Enterprise Business Intelligence	Sales Data Mart, HR Data Mart

Application of Data Warehouse

Real World Implementations

Application Area	Purpose of Data Warehouse
Banking	Fraud detection, loan analysis, customer profitability
Retail & E-Commerce	Sales analysis, recommendations, inventory management
Healthcare	Patient history, disease analysis, resource planning
Education	Student performance, admissions, placements
Telecommunications	Customer churn, network analysis, usage patterns
Manufacturing	Production planning, quality control, supply chain analysis
Insurance	Claim analysis, fraud detection, risk assessment
Government	Census, taxation, public policy, budgeting
Airlines	Passenger trends, route optimization, revenue analysis
Marketing & CRM	Customer segmentation, campaign analysis
Business Intelligence	Dashboards, KPIs, executive reporting
Data Mining	Pattern discovery, prediction, recommendations
Inventory Management	Demand forecasting, stock optimization
Human Resources	Employee analytics, recruitment, attrition analysis

References

- Primary References
- Anahory, S., & Murray, D. (1997). Data Warehousing in the Real World: A Practical Guide for Building Decision Support Systems. Addison-Wesley Professional.
- Ponniah, P. (2010). Data Warehouse Fundamentals: A Comprehensive Guide for IT Professionals (2nd ed.). Wiley India Pvt. Ltd.
- Course Presentation (PPT) provided by the Department of Information and Communication Technology (ICT), used as reference material for the Data Warehousing course.

References

- Additional Standard References
- Inmon, W. H. (2005). Building the Data Warehouse (4th ed.). Wiley.
- Kimball, R., & Ross, M. (2013). The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling (3rd ed.). Wiley.
- Chaudhuri, S., & Dayal, U. (1997). "An Overview of Data Warehousing and OLAP Technology." ACM SIGMOD Record, 26(1), 65–74.
- **CHATGPT FOR CONTENT MANAGEMENT**

Thank you

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